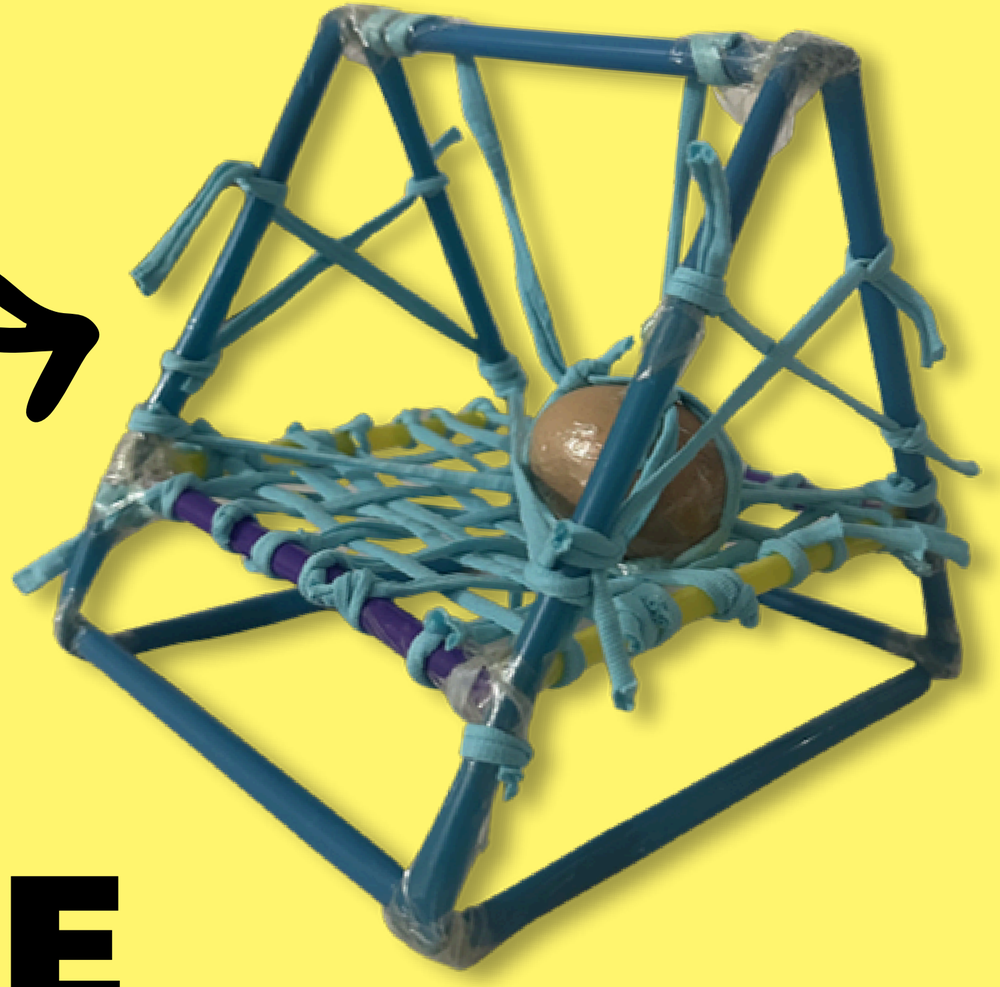
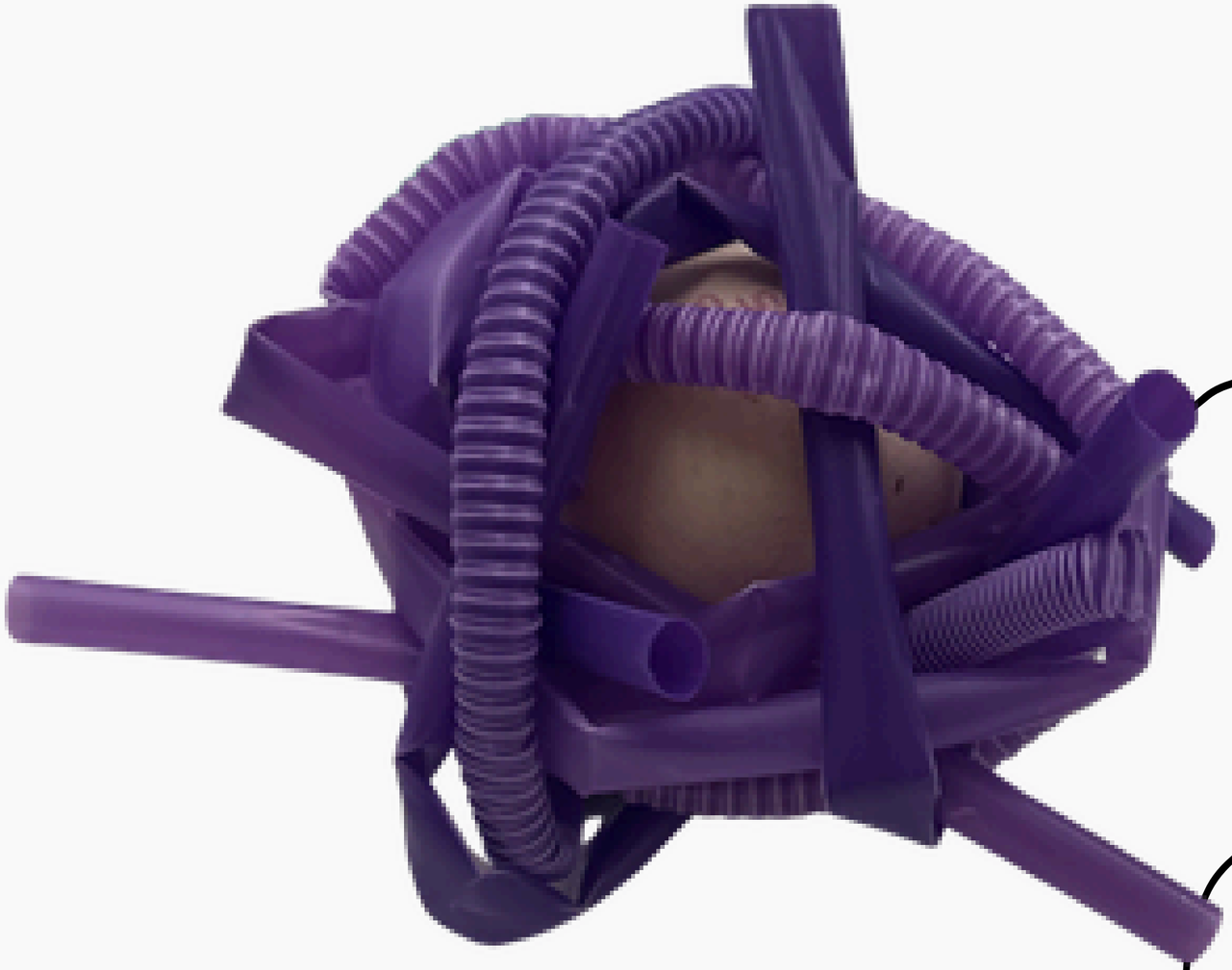
IDE102 X AI 26

EGG DROP CHALLENGE



BEGUM
HACICAFEROGLU



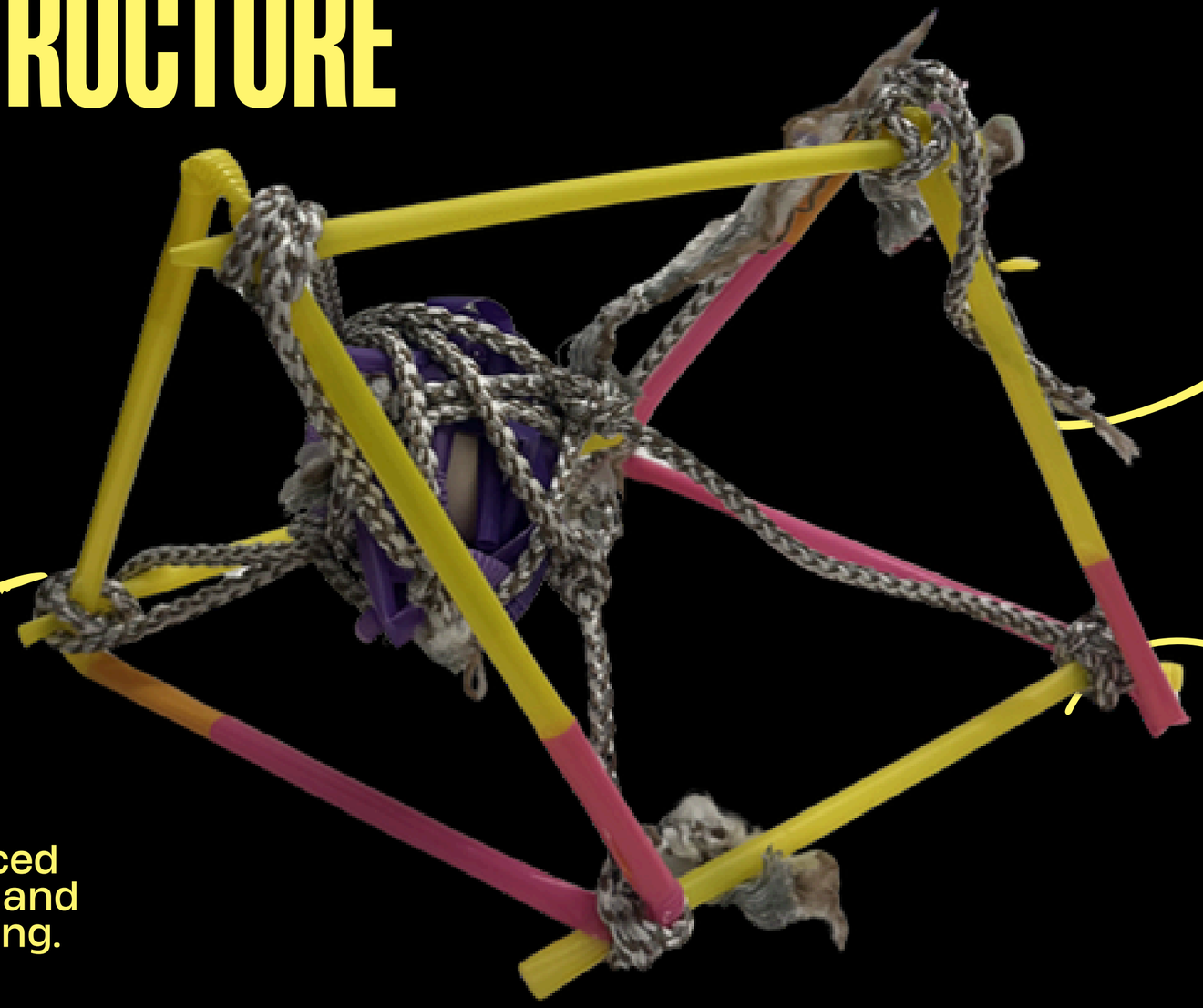
DIRECT PROTECTION

My first instinct was to directly protect the egg.

I wrapped the egg with small straw pieces to create a protective outer shell.

The idea was that the straw layer would act as a cushion and prevent the egg from cracking during the impact.

FIRST STRUCTURE



After protecting the egg, I built a triangular prism structure using straws.

The egg was placed inside the frame and secured with string.

Triangular geometry was chosen because triangles are structurally stable and resist deformation.

FIRST DROP TEST

The egg survived the drop, but the structure did not behave as expected.

The frame bent and lost its symmetry during impact, causing the system to tilt instead of landing vertically.

After analyzing the result, ı realized that the egg survived mainly because of the straw shell wrapped around it, not because of the structural frame.

The real challenge was not simply protecting the egg, but controlling how the impact energy is distributed during the fall.

This showed that the design was relying on direct protection rather than stryctural logic.

CONSULTING AI

PROMPT 1

“How can I design a structure that prevents an egg from breaking during a 3 meter drop using straws and string?„

AI explained that the problem is related to energy dissipation.

The egg should not stop instantly after the fall.

Instead, the structure should slow the egg down gradually.

ENERGY DISSIPATION

If the object stops suddenly, all that energy is transferred instantly.

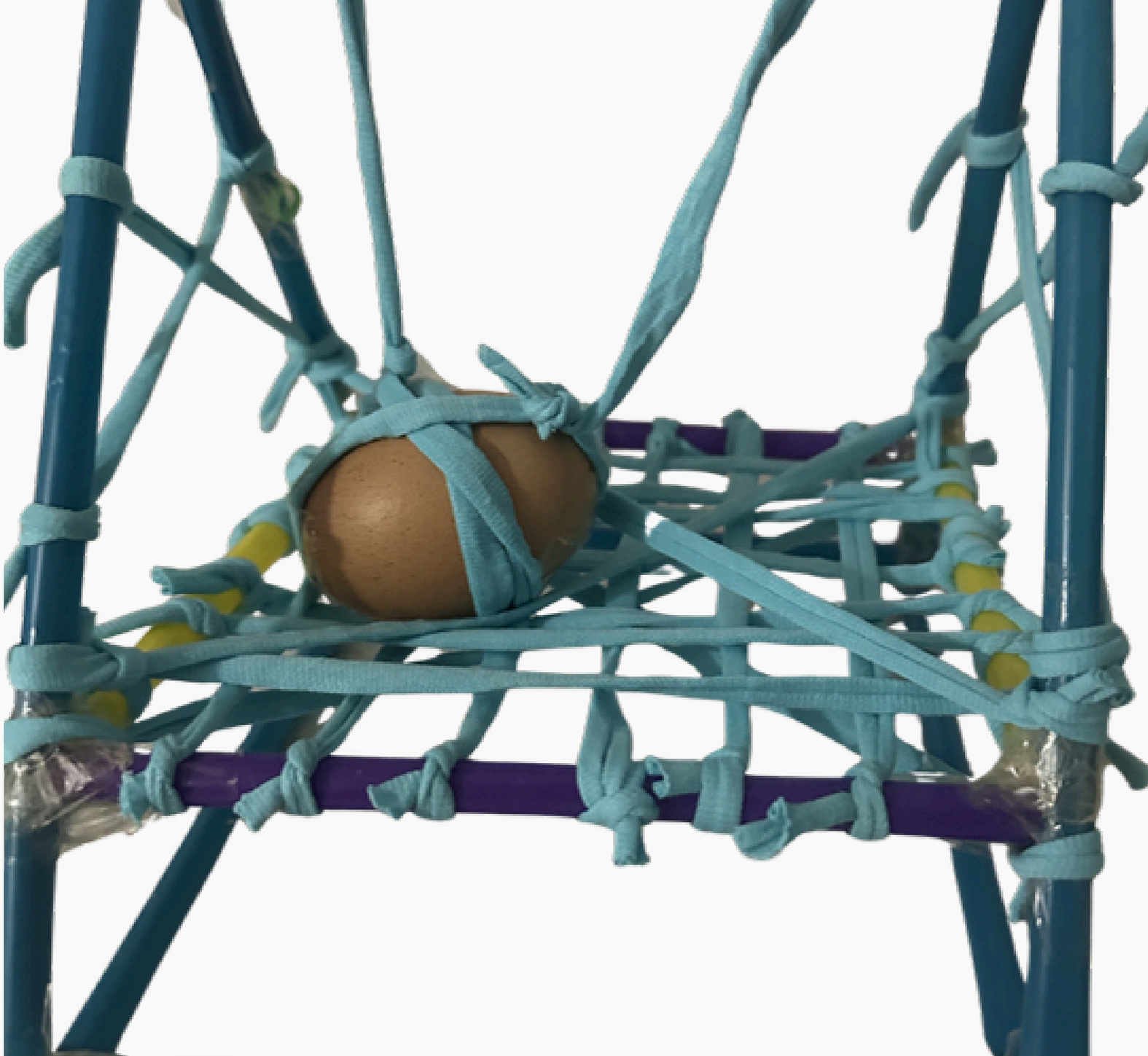
When an object falls, it gains kinetic energy.

This sudden transfer creates a large force that can break the egg.

AI suggested allowing a small movement so that the energy is spread over time.

SUSPENSION SYSTEM

Inspired by this idea, I designed a string suspension system. Instead of fixing the egg rigidly inside the frame, I placed it in a small string hammock.



CONTROLLING THE FALL

The movement helped absorb and distribute the force.

FINAL STRUCTURE LOGIC

01 LOW CENTER OF GRAVITY → The egg is positioned lower in the frame to increase stability.

02 TENSION SUSPENION → The egg is supported by a string network that acts like a hammock.

03 ENERGY DISSIPATION → When the structure hits the ground, the egg continues moving slightly before stopping, reducing peak impact force.

REFLECTION

AI played an important role in reframing the problem.
While the first prototype focused on direct protection, AI introduced the idea of energy dissipation and suspension systems.

However, physical testing remained essential.
Some AI suggestions were conceptually useful but required simplification to become buildable.

The final design emerged from a feedback loop between AI suggestions and physical experimentation.

Good design is not only about building stronger structures but about understanding how forces move through a system.